

RFC 7523 and OA4MP

This is a quick note about how to get a client using RFC 7523 under OA4MP. The aim is to get a developer who needs this to work now up to speed.

RFC:<https://www.rfc-editor.org/rfc/rfc7523>

Executive Summary

Register your client with a public key and you can just send requests (as a signed JWT) directly for access tokens – no user involved. This is § 2.1 of the spec. This is far better than that client credential flow – none of the vulnerabilities – and OA4MP supports it right now.

You can alternately just use your public key in place of a client password . This is §2.2 of the spec.

OA4MP requires you get authorized as per §2.2 to make the request as per §2.1, so we need both of these. We may relax that in the future. May.

I'll stick to an annotated example of how to make the request to the token endpoint. The initial kickoff for the flow is RFC 7523, but the flow after that is identical to the device or auth code flow.

Registration with OA4MP

If you use the registration endpoint (oauth2/register) then there is a box for the public key. You put your public keys in JWK (JSON webkey) format. This can include multiple keys with identifiers (**kid**). You keep the private keys and sign your request JWTs with them. Note that you can use a single key, in which case no kid is required in the request, but if you have multiple, you must include it.

(**Nota Bene:** This is for standard OA4MP. For CILogon (which is an extension of OA4MP), this feature is disabled and only available to paying subscribers .)

If you are creating or approving the client as an system administrator, set the **service_client** property to true using the CLI.

Using the Client Management Endpoint.

If you have an administrative client for OA4MP, you can simply upload your public key(s). Remember that you can have multiple public keys. This is documented in [RFC 7591](#) section 2, under client metadata. You need to supply the **token_endpoint_auth_method** with a value of **private_key_jwt** as per RFC 7523. Then you supply either **jwks_uri** or **jwks**. These, by the way are exclusive. The spec is clear that if you can use a JWKS uri (which is just your JWKS) then you must, since it allows for you to control such things as key rotations. Failing that, you can just upload them

Example. Using a jwks_uri

In this example snippet (this is just a tiny part of your full registration), you give the coordinates for your public key(s) only:

```
{
  "token_endpoint_auth_method": "private_key_jwt",
  "jwks_uri": "https://keys.bigstate.edu/public\_keys/bob.jwks",
  "client_name": "My Example Client", ... lots more
}
```

Example. Uploading the JSON web keys directly.

In this example, there is one key (note that the format still must be for a full set of them, which means you send an array).

```
{
  "token_endpoint_auth_method": "private_key_jwt",
  "jwks": {
    "jwk": [
      {
        "alg": "EC",
        "crv": "P-256",
        "x": "MKBCTNICKUSDi...",
        (lots more!!)
      }
    ]
  },
  "client_name": "My Example Client", ... lots more
}
```

Again, if you upload both `jwks_uri` and `jwks` then as per the specification, the server will reject the request. If this works, then your client is now RFC 7523 capable and you can use it as described in the rest of this document, for either just authorization or to create signed authorization grants. Note that you choose which key is used by specifying the kid (see below) when sending a request to the server.

Authorization (authentication, actually) (§2.2)

A typical request to the token endpoint POSTs the following two things

```
client_assertion_type=urn:iETF:params:oauth:client-assertion-type:jwt-bearer
client_assertion=signed auth JWT
```

The **client_assertion_type** is fixed and required. The "signed auth JWT" is a JWT that is signed with your private key. OA4MP uses the stored public key to verify.
Typical example is (line breaks added, signature truncated)

eyJraWwQI0iJCMzNGODZBMzI3QTizMkU5IiwidHlwIjoiSldUIiwiaWxnIjoiUlMyNTYifQ
 .eyJhdWQiOiJodHRwciovL2xvY2FsaG9zdDo5NDQzL29hdXRoMi90b2t1biIsImV4cCI6MTcxNzYyYmJyYmMC
 wiaWF0IjojOjE3NjIxNzUwLCJpc3MiOiJhdXRvLXRoIiwic3Q6L29hdXRvL3JmYzkwNjgvcWRsIiwianRpIjoiY
 XV0by10ZXN0IiwiaGVhZC9yZWMDY4L3FkbC9yZWMDNTIzL0ZIZlRab042WHdwSjk5Y1ZaMGFETWJpTXg1
 Z2xHTG1EY1hHSWtyTUM2UmMiInN1YiI6ImF1dG8tdGVzdDovb2F1dGdvcmZjOTA2OC9xZGwiLH0

.ukG2PD8_InSZ8cdFn1FHs_tSv_cpFUEIGMfv-0y2xXDNkk73NM1sIuBfgSLS_0v...

which has header

```
{
  "kid": "B33F86A327A232E9",
  "typ": "JWT",
  "alg": "RS256"
}
```

Key	Description
kid	The key identifier in the JWK used at registration.
typ	Type of payload. This is fixed at JWT
alg	The algorithm used. Required

Again, if you uploaded a single key, you don't need a kid. The payload decodes as

```
{
  "aud": "https://localhost:9443/oauth2/token",
  "exp": 1717622650,
  "iat": 1717621750,
  "iss": "auto-test:/oauth/rfc9068/qdl",
  "jti":
"auto-test:/oauth/rfc9068/qdl/rfc7523/FHFTZoN6XwpJ99cVZ0aDMbIMx5glGLmDcXGIkrMC6Rc"
  "sub": "auto-test:/oauth/rfc9068/qdl"
}
```

Notes

key	Req?	Description
aud	Y	The server address
exp	Y	Timestamp in seconds when this request expires
iat	N	Timestamp in seconds when this request was created
iss	Y	The client identifier
jti	N	An identifier created by the client. This is ignored by the server but is passed back at times.
sub	Y	The client identifier. Fixed, as per spec. Same as iss here.

Authorization Grant (§2.1)

Quick review: A standard authorization code flow sends a request, the user needs to authenticate, then the client gets a response which includes the authorization grant. This response is used at the token endpoint to get an access token.

RFC 7523 lets you simply write your own authorization grant and send it to the token endpoint. The trick, of course, is that your client has a trust relationship with the server (the public key) so we know it can only have come from a specific client and cannot be forged.

The request POSTs the following

```
grant_type=urn:ietf:params:oauth:grant-type:jwt-bearer
assertion=auth grant JWT
```

The **grant_type** is fixed and must be as given. The auth grant JWT is a JWT that is signed with your private key. OA4MP uses the stored public key to verify. Typical example is (line breaks added, signature truncated)

```
eyJraWQiOiJCMzNGODZBMzI3QTIZMkU5IiwidHlwIjoiSldUIiwiaWxnIjoiUlMyNTYifQ
.eyJhdWQiOiJodHRwczovL2xvY2FsaG9zdDo5NDQzL29hdXRoMi90b2t1biIsImV4cCI6MTcxNzYyMjY1MC
wiaWF0IjoxNzE3NjI1NjI3MjY1IiwiaWF0IjoxNzE3NjI1NjI3MjY1IiwiaWF0IjoxNzE3NjI1NjI3MjY1
XV0by10ZXN0IiwiaWF0IjoxNzE3NjI1NjI3MjY1IiwiaWF0IjoxNzE3NjI1NjI3MjY1IiwiaWF0IjoxNzE3
aVNWbW9wNjY1IiwiaWF0IjoxNzE3NjI1NjI3MjY1IiwiaWF0IjoxNzE3NjI1NjI3MjY1IiwiaWF0IjoxNzE3
VbW9wNjY1IiwiaWF0IjoxNzE3NjI1NjI3MjY1IiwiaWF0IjoxNzE3NjI1NjI3MjY1IiwiaWF0IjoxNzE3
luZm8iXSwic3ViIjoiZGF2ZW5wb3J0In0
.orJxMrEbQa-Q7Mue...
```

The header decodes as

```
{
  "kid": "B33F86A327A232E9",
  "typ": "JWT",
  "alg": "RS256"
}
```

See above, since this is identical. The payload decodes as

```
{
  "aud": "https://localhost:9443/oauth2/token",
  "exp": 1717622650,
  "iat": 1717621750,
  "iss": "auto-test:/oauth/rfc9068/qdl",
  "jti": "auto-test:/oauth/rfc9068/qdl/rfc7523/N8J05Uop7_-ocRGY_XthqTXzw-
iSV7HQbeDPe3kAPms",
  "nonce": "utrVjhgfFgEx4yKbtP43GSpvbFmmpjZDtxcUn0i2ANK",
  "resource": "ANY",
  "scope": ["openid", "org.cilogon.userinfo"],
  "sub": "davenport"
}
```

Note that any parameters you can normally send in a request should be encoded here – this is the actual request, not other parameters. Hence the scope and resource.

Key	Req ?	Description
aud	Y	The address of the server
exp	Y	The expiration timestamp, in seconds

iat	N	The issued at timestamp, in seconds
id_token	N	A JSON object that is the ID token. This will be returned in e.g. the user info endpoint.
iss	Y	The client identifier
jti	N	A client generated identifier
nonce	N	A nonce (one-time random string) a client may use to track requests
oa4mp:/jwt/signing/keyID		ID for signing keys. These are found on the well-known page in the signing keys. This will be used for all token signing (unless you request a different one). Not sending it means to let the server use its default.
resource	N	OA4MP request parameter
scope	N	OA4MP request parameter
state	N	The state parameter. This is merely echoed back as per specification.
sub	Y	Usually the user name. This will be used as the subject of the identity token. If sending the id_token, it may be included in that.

The **id_token** is normally generated by the system during the authorization but since you are controlling that with this request, only the most basic will be made. Sending one that asserts claims such as user groups, an identity provider etc. means that will be taken as the basic id token.

RFC7523 and the device flow (RFC 8628)

You may register your client's key(s) and simply use them for authorization (§2.2) in the device code flow.

```
{
  "alg": "RS256",
  "kid": "EC9FCFCB3716AC4C2279DF42EC98CABF",
  "typ": "JWT"
}
```

The only slight change is to note that, as per the spec., the aud claim is the endpoint, so it must be the device code flow endpoint:

```
{
  "aud": "https://localhost:9443/oauth2/device_authorization",
  "exp": 1717709668,
  "iat": 1717708768,
  "iss": "ashigaru:command.line2",
  "jti":
"ashigaru:command.line2/rfc7523/SpPddDaTx6SFJ7Cd9x4DYbwhQ4RtaM6FWSLUU3KJJr4"
  "sub": "ashigaru:command.line2",
}
```